# File I/O

SKK UNIVERSITY

# File & Directory

# What is File?

- **A Linux file is a sequence of *m* bytes**
  - $B_0, B_1, \ldots, B_k, \ldots, B_{m-1}$

- **All I/O devices are represented as files**
  - **`/dev/sda1`** (hard disk partition)
  - **`/dev/tty2`** (terminal)
    - » Ctrl + Alt + F1 ~ F7

- **Even the kernel is represented as a file**
  - **`/boot/vmlinux-6.8.0-55-generic`** (kernel image)
  - **`/dev/mem`** (kernel memory image)
  - **`/proc`** (kernel data structures)

# File Types

- **Each file has a *type* indicating its role in the system**
  - **Regular file** : contains arbitrary data
  - **Directory** : index for a related group of files
  - **Socket** : for communicating with a process on another machine
  - **Named pipe** : for communicating with another process

- **Other file types beyond our scope**
  - Symbolic links
  - Character and block devices - `/dev/null`, `/dev/zero`

# Regular File

- **A regular file contains arbitrary data**

- **Applications often distinguish between *text files* and *binary files***
  - Text files are regular files with only ASCII or Unicode characters
  - Binary files are everything else
    - » e.g. object files, JPEG images
  - Kernel doesn't distinguish the two during I/O operations.

- **Text file is sequence of text lines**
  - Text line is sequence of *chars* terminated by newline *char* (`'\n'`)

# Directory

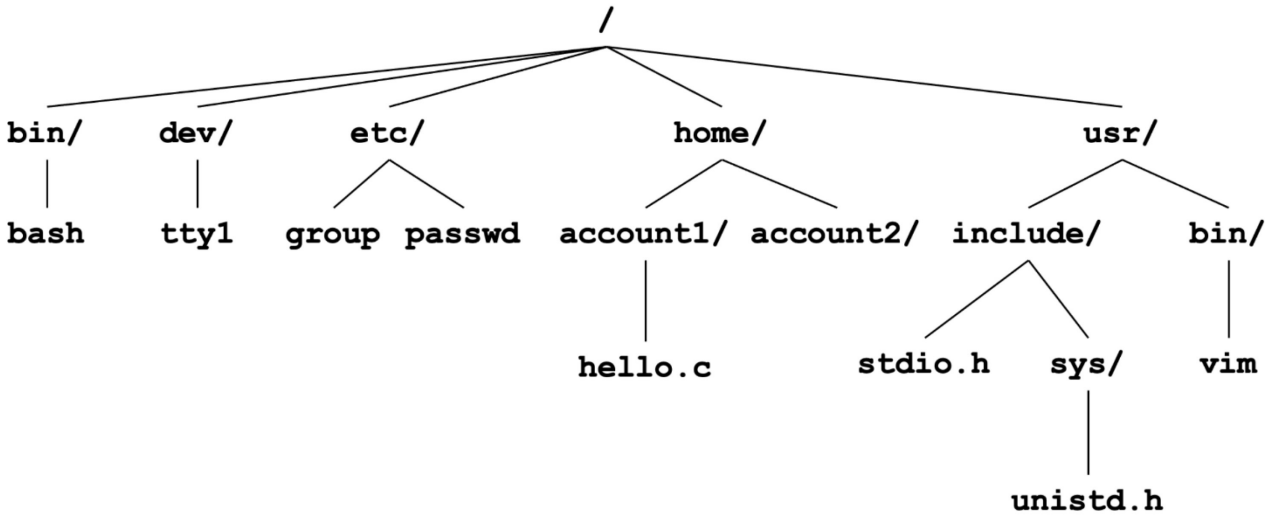
- **Directory consists of an array of *links***
  - Each link maps a *filename* to a file

- **Each directory contains at least two entries**
  - `.` (single dot) is a link to itself
  - `..` (double dot) is a link to the parent directory in the directory hierarchy

- **Commands for manipulating directories**
  - **`mkdir`** : create empty directory
  - **`ls`** : view directory contents
  - **`rmdir`** : delete empty directory

# Directory Hierarchy

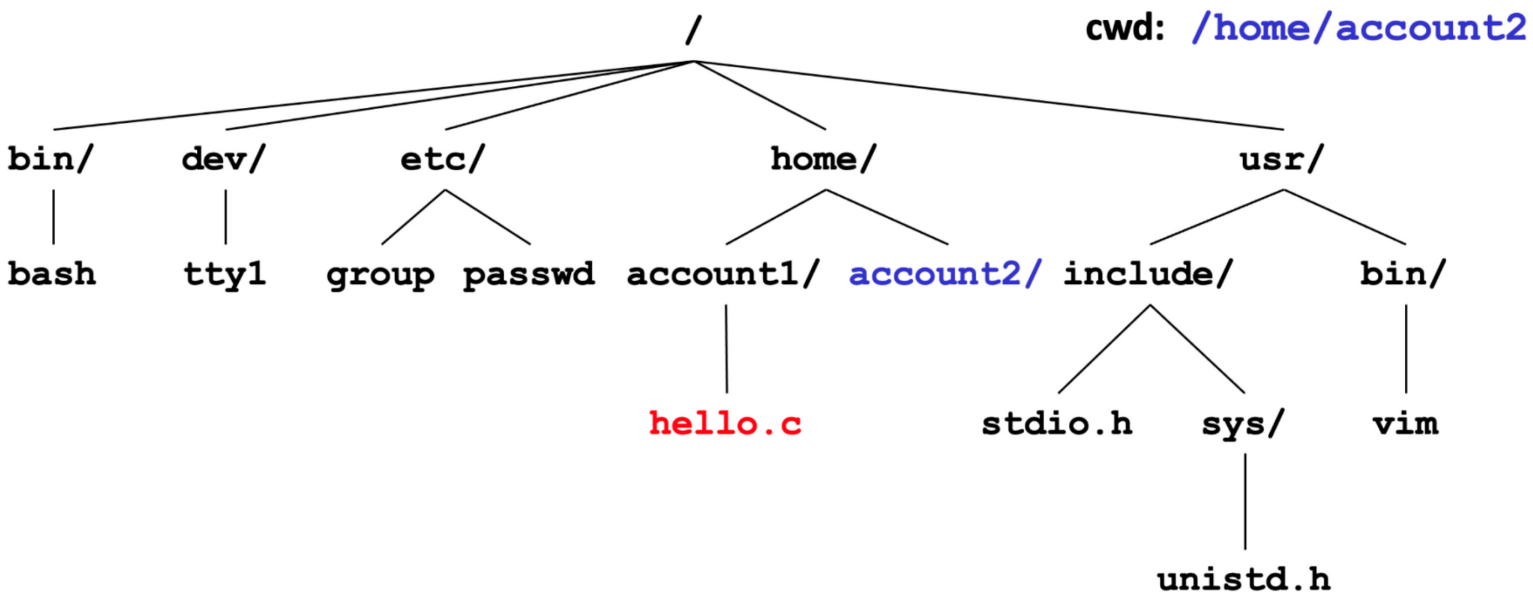
- **All files are organized as a hierarchy anchored by root directory named '/' (slash)**




- **Kernel maintains *current working directory* (cwd) for each process**
  - Modified using the `cd` command

# Pathnames

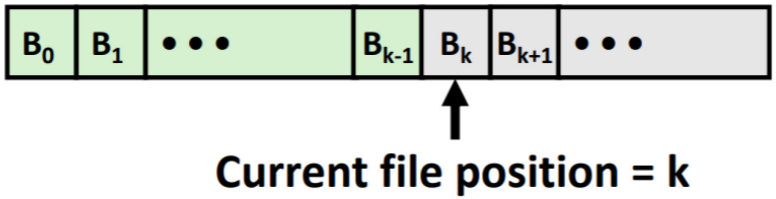
- **Locations of files in the hierarchy denoted by *pathnames***
  - *Absolute pathname* starts with '/' and denotes path from *root*
    - » `/home/account1/hello.c`
  - *Relative pathname* denotes path from *current working directory*
    - » `../account1/hello.c`



# Unix I/O

# Unix I/O Overview

- **Kernel exports a simple interface called *Unix I/O***
  - Opening and closing files
    - » `open()` and `close()`
  - Reading and writing a file
    - » `read()` and `write()`
  - Changing the current file position (seek)
    - » Indicates next offset into file to read or write
    - » `lseek()`




- **All input and output is handled in a consistent and uniform way ("*byte stream*")**

# Opening Files

- **Opening a file informs the kernel that you are getting ready to access that file**

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
```

```
int fd;    /* file descriptor */

if ((fd = open("/etc/hosts", O_RDONLY)) < 0) {
    perror("open");
    exit(1);
}
```

- **Returns a small identifying integer *file descriptor***
  - **`fd == -1`** indicates that an error occurred

- **Each process created by a Unix shell begins life with three open files associated with a terminal**
  - 0 (**`STDIN_FILENO`**): standard input (**`stdin`**)
  - 1 (**`STDOUT_FILENO`**): standard output (**`stdout`**)
  - 2 (**`STDERR_FILENO`**): standard error (**`stderr`**)

# Opening File Flags

- **Flags for open( ) function**
  - `O_RDONLY` : open for reading file
  - `O_WRONLY` : open for writing file
  - `O_RDWR` : open for reading and writing
  - `O_CREAT` : create the file if it doesn't exist
  - `O_APPEND` : append to the end of file

- **When creating the file (with O_CREAT flag), 3rd argument *mode* must be supplied**

```
fd = open("c.txt", O_CREAT, 0764);
```

  - **0764** equals **S_IRWXU | S_IRGRP | S_IWGRP | S_IROTH**

| Permissions | Size | User | Group | Date Modified | Name |
|---|---|---|---|---|---|
| .rw-rw-r-- | 0 | spl | spl | 22 Mar 22:15 | week4.txt |

# Review: File Permissions

- **Typing `ls -l` allows you to check the permissions for file(s)**

```
Permissions Size User Group Date Modified Name
 rw-rw-r--    0 spl  spl   22 Mar 22:15  week4.txt
```

owner group others

- **Three types of permission: `read` (+4), `write` (+2), and `execute` (+1)**
- **Three types of people:**
  - owner (u): the user who owns the file. By default, the user who created the file, but can be changed using chown.
  - group (g): the group associated with the file.
  - others (o): everyone else
- **Octal code represents the sum of permissions for each type of people (i.e. 0664 in the example)**
- **You can also use enum values (e.g. S_IRWXU, S_IRGRP, S_IROTH)**
  - Check “man 2 open” for all the flags under the O_CREAT option

# Closing Files

- **Closing a file informs the kernel that you are finished accessing that file**

```
int fd = ...;      /* file descriptor */
int retval; /* return value */

if ((retval = close(fd)) < 0) {
    perror("close");
    exit(1);
}
```

- **Closing an already closed file is a recipe for disaster in threaded programs (more on this later)**

- **Moral: Always check return codes, even for seemingly benign functions such as `close()`**

# Reading Files

- **Reading a file copies bytes from the *current file position* to *memory*, and then updates file position**

```
char buf[512];
int fd = ...;        /* file descriptor */
size_t nbytes;    /* number of bytes read */
/* Open file fd ...   */
/* Then read up to 512 bytes from file fd */
if ((nbytes = read(fd, buf, sizeof(buf))) < 0) {
    perror("read");
    exit(1);
}
```

- **Returns number of bytes read from file `fd` into `buf`**
  - **`nbytes < 0`** indicates that an error occurred
  - Short counts are possible and are not errors! (**`nbytes < sizeof(buf)`**)

# Writing Files

- **Writing a file copies bytes from *memory* to the *current file position*, then updates current file position**

```
char buf[512] = ...;
int fd = ...;        /* file descriptor */
size_t nbytes;    /* number of bytes read */

/* Open the file fd ... */
/* Then write up to 512 bytes from buf to file fd */
if ((nbytes = write(fd, buf, sizeof(buf))) < 0) {
    perror("write");
    exit(1);
}
```

- **Returns number of bytes written from `buf` to file `fd`**
  - **`nbytes < 0`** indicates that an error occurred
  - As with reads, short counts are possible and are not errors!

# File Offset

- **An *offset* of an opened file can be set explicitly by calling `lseek()`, `lseek64()`**

```
char buf[512] = ...;
int fd = ...;        /* file descriptor */
off_t pos;     /* file offset */

/* Get current file offset */
pos = lseek(fd, 0, SEEK_CUR);
/* The file offset is incremented by written bytes */
write(fd, buf, sizeof(buf));
/* Set file position to the first byte of the file */
pos = lseek(fd, 0, SEEK_SET);
```

- **Returns the new *offset* of the file `fd`**
  - **`offset < 0`** indicates that an error occurred
  - An offset can be set beyond the end of the file
    - » If data is written at that point, a file "hole" is created

# Simple Unix I/O Example

- Copying standard input to standard output one byte at a time

```
#include <unistd.h>

int main(void) {
    char c;

    while(read(STDIN_FILENO, &c, 1) != 0)
        write(STDOUT_FILENO, &c, 1);

    return 0;
}
```

- Can terminate with End-of-Transmission character (Ctrl+D in Unix-like systems)

# Short Counts

- ***Short counts* can occur in these situations:**
  - Encountering (end-of-file) EOF on reads
  - Reading text lines from a terminal
  - Reading and writing network sockets

- ***Short counts* does not occur in these situations:**
  - Reading from disk files (except for EOF)
  - Writing to disk files

- **Best practice is to always allow for short counts**

# Handling Short Counts Example

```
#include <unistd.h>
#include <errno.h>

ssize_t rio_readn(int fd, void *usrbuf, size_t n) {
  size_t nleft = n;
  ssize_t nread;
  char *bufp = usrbuf;

  while (nleft > 0) {
    if ((nread = read(fd, bufp, nleft)) < 0) {
      if (errno == EINTR) /* interrupted by signal handler return */
        nread = 0; /* and call read() again */
      else
        return -1; /* errno set by read() */
    } else if (nread == 0) break; /* EOF */
    nleft -= nread;
    bufp += nread;
  }

  return (ssize_t) (n - nleft); /* return >= 0 */
}
```

# File Metadata

- **Metadata is data about data, in this case file data**
  - Per-file metadata maintained by kernel, accessed by users with the **`stat`** and **`fstat`** functions

```
/* Metadata returned by the stat and fstat functions */
struct stat {
    dev_t         st_dev;     /* device */
    ino_t         st_ino;     /* inode */
    mode_t        st_mode;    /* protection and file type */
    nlink_t       st_nlink;   /* number of hard links */
    uid_t         st_uid;     /* user ID of owner */
    gid_t         st_gid;     /* group ID of owner */
    dev_t         st_rdev;    /* device type (if inode device) */
    off_t         st_size;    /* total size, in bytes */
    unsigned long st_blksize; /* blocksize for filesystem I/O */
    unsigned long st_blocks;  /* number of blocks allocated */
    time_t        st_atime;   /* time of last file access */
    time_t        st_mtime;   /* time of last file modification */
    time_t        st_ctime;   /* time of last inode change */
};                            /* statbuf.h included by sys/stat.h */
```

# System Calls for File Metadata

- **An information about a file can be obtained explicitly by calling `stat()`, `fstat()`, `lstat()`**
  - Three *header files* must be included

```
#include <sys/types.h>
#include <sys/stat.h>
#include <unistd.h>
```

  - **`int stat(const char *path, struct stat *buf)`**
    - » Fills file metadata in **`buf`** pointed by path
  - **`int fstat(int fd, struct stat *buf)`**
    - » Identical to **`stat()`**, except that the file is specified by the file descriptor **`fd`**
  - **`int lstat(const char *path, struct stat *buf)`**
    - » Identical to **`stat()`**, except that if path is a symbolic link, fill link itself, not the file

# Ex) Accessing File Metadata

```c
/* statcheck.c - Querying and manipulating a file's meta data */

#include <sys/stat.h>
#include <stdio.h>
#include <stdlib.h>

int main (int argc, char **argv) {
    struct stat st;
    char *type, *readok;

    stat(argv[1], &st);
    if (S_ISREG(st.st_mode)) /* Determine file type */
        type = "regular";
    else if (S_ISDIR(st.st_mode))
        type = "directory";
    else
        type = "other";
    if ((st.st_mode & S_IRUSR)) /* Check read access */
        readok = "yes";
    else
        readok = "no";

    printf("type: %s, read: %s\n", type, readok);
    exit(0);
}
```

```
$ ./statcheck statcheck.c
type: regular, read: yes
$ chmod 000 statcheck.c
$ ./statcheck statcheck.c
type: regular, read: no
```

# Standard I/O

# Standard I/O Functions

- **The C standard library (`libc.so`) contains a collection of higher-level *standard I/O* functions**

- **Examples of standard I/O functions**
  - Opening and closing files (**`fopen()`** and **`fclose()`**)
  - Reading and writing bytes (**`fread()`** and **`fwrite()`**)
  - Reading and writing text lines (**`fgets()`** and **`fputs()`**)
  - Formatted reading and writing (**`fscanf()`** and **`fprintf()`**)
  - Opening and closing directories (**`opendir()`** and **`closedir()`**)
  - Reading directory's entries (**`readdir()`**)

# Standard I/O Streams

- **Standard I/O models open files as *streams***
  - Abstraction for a *file descriptor* and a *buffer* in memory

- **C programs begin life with three open streams (defined in `stdio.h`)**
  - **`stdin`** (standard input)
  - **`stdout`** (standard output)
  - **`stderr`** (standard error)

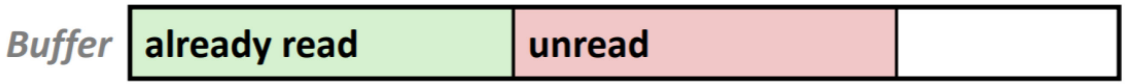
```
#include <stdio.h>
extern FILE *stdin;  /* standard input  (descriptor 0) */
extern FILE *stdout; /* standard output (descriptor 1) */
extern FILE *stderr; /* standard error  (descriptor 2) */

int main() {
    fprintf(stdout, "Hello, world\n");
}
```

# Buffered I/O : Motivation

- **Applications often read/write one character or line at a time**
  - **`getc(), putc(), ungetc()`**
  - **`gets(), fgets()`**
    - » Read line of text one character at a time, *stopping at newline*

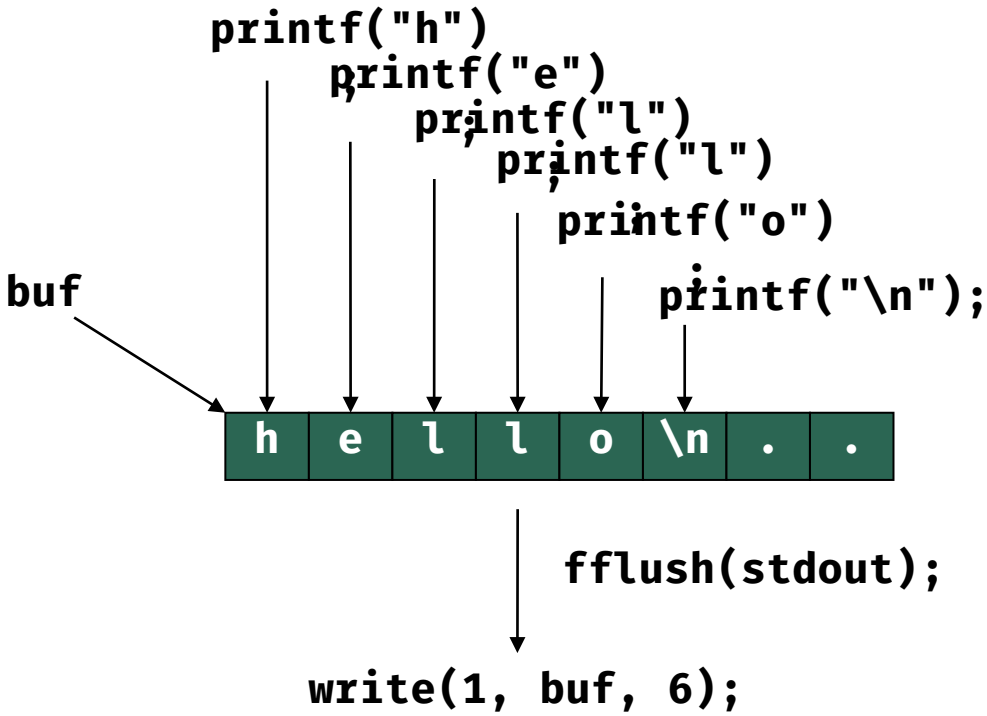
- **Implementing as Unix I/O calls expensive**
  - Each **`read()`** and **`write()`** require Unix kernel calls
    - » > 1,000 clock cycles for simple system calls

- **Solution : buffered read**
  - Use Unix **`read()`** to grab block of bytes
  - User input functions take one byte at a time from buffer
    - » Refill buffer when empty



# Buffering in Standard I/O (1)

- Standard I/O functions use buffered I/O




- Buffer flushed to output `fd` on `"\n"`, call to `fflush()` or `exit()`, or return from main

# Buffering in Standard I/O (2)

- You can see this buffering in action for yourself, by using Linux program, `strace`.

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    printf("h");
    printf("e");
    printf("l");
    printf("l");
    printf("o");
    printf("\n");
    fflush(stdout);
    exit(0);
}
```

```
linux> strace ./hello
execve("./hello", ["hello"], [/* ... */]).
...
write(1, "hello\n", 6)                  = 6
...
exit_group(0)                           = ?
```

# Accessing Directory

- **Only recommended operation on a directory: read its *entries***
  - **`dirent`** structure contains information about a directory entry

```
struct dirent {
    ino_t          d_ino;       /* inode number */
    off_t          d_off;       /* offset to the next dirent */
    unsigned short d_reclen;    /* length of this record */
    unsigned char  d_type;      /* type of file */
    char           d_name[256]; /* filename */
};
```

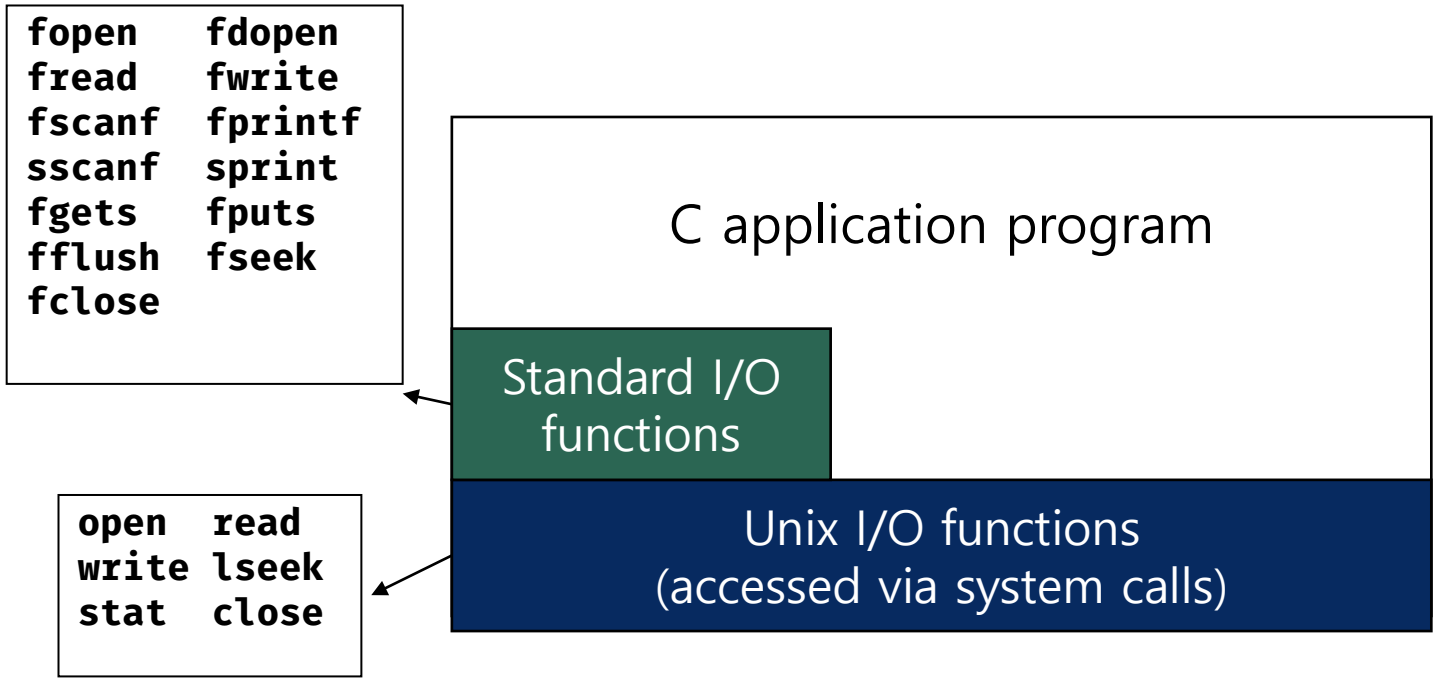
  - **`DIR`** structure contains information about directory while stepping through its entries (see example in the next slide)
  - **`opendir()`** and **`closedir()`**
    - » Open and close directory stream
  - **`readdir()`**
    - » Return a pointer to a **`dirent`** structure representing directory entry
    - » Return **`NULL`** on reaching the end of the directory stream
    - » Unix I/O syscall equivalent: getdents64 (but no C function).

# Ex) Accessing Directory

```c
/* dircheck.c – Opening directory and reading its entries
Compile with --std=gnu2x or --std=gnu23*/

#include <sys/types.h>
#include <dirent.h>
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char **argv) {
    DIR* directory;
    struct dirent *de;

    if (!(directory = opendir(argv[1]))) {
        perror("Failed to open directory");
        exit(1);
    }

    while ((de = readdir(directory)) != nullptr)
        printf("Found: %s\n", de->d_name);

    closedir(directory);
    exit(0);
}
```

# Unix I/O vs. Standard I/O

- **Standard I/O are implemented with low-level Unix I/O**




- **Which ones should you use in your programs?**

# Pros and Cons of Unix I/O

- **Pros**
  - The most general and lowest overhead form of I/O
    - » All other I/O packages are implemented using Unix I/O functions
  - Unix I/O provides functions for accessing *file metadata*

- **Cons**
  - System call overheads for small-sized I/O
  - Dealing with short counts is tricky and error prone
  - Efficient reading of text lines requires some form of buffering, also tricky and error prone
  - These issues are addressed by the standard I/O

# Pros and Cons of Standard I/O

- **Pros**
  - Buffering increases efficiency by decreasing the number of **`read()`** and **`write()`** system calls
  - Short counts are handled automatically

- **Cons**
  - Provides no function for accessing file metadata
  - Standard I/O is not appropriate for input and output on network sockets
    - » But there is a way using **`fdopen()`**

## Error Handling for

# File I/O

# perror Function / errno Variable

- **When a system call fails,**
  - Returns `-1` (or `NULL` for certain library functions)
  - The latest error information is stored in "`errno`"

- **Function `strerror`**
  - Returns a pointer to a string explaining error information stored in `errno`

- **Function `perror`**
  - Prints out a description of the error stored in `errno` through `stderr` stream.

- **Variable `errno`**
  - Stores `int` value indicating cause of error
  - int type extern global variable
  - Thread-safe!

# errno.h

- **/usr/include/asm-generic/errno-base.h**

File: **/usr/include/asm-generic/errno-base.h**

```
/* SPDX-License-Identifier: GPL-2.0 WITH Linux-syscall-note */
#ifndef _ASM_GENERIC_ERRNO_BASE_H
#define _ASM_GENERIC_ERRNO_BASE_H

#define EPERM        1  /* Operation not permitted */
#define ENOENT       2  /* No such file or directory */
#define ESRCH        3  /* No such process */
#define EINTR        4  /* Interrupted system call */
#define EIO      5  /* I/O error */
#define ENXIO        6  /* No such device or address */
#define E2BIG        7  /* Argument list too long */
#define ENOEXEC      8  /* Exec format error */
#define EBADF        9  /* Bad file number */
#define ECHILD      10  /* No child processes */
#define EAGAIN      11  /* Try again */
#define ENOMEM      12  /* Out of memory */
#define EACCES      13  /* Permission denied */
#define EFAULT      14  /* Bad address */
#define ENOTBLK     15  /* Block device required */
#define EBUSY       16  /* Device or resource busy */
#define EEXIST      17  /* File exists */
#define EXDEV       18  /* Cross-device link */
#define ENODEV      19  /* No such device */
#define ENOTDIR     20  /* Not a directory */
#define EISDIR      21  /* Is a directory */
#define EINVAL      22  /* Invalid argument */
#define ENFILE      23  /* File table overflow */
#define EMFILE      24  /* Too many open files */
#define ENOTTY      25  /* Not a typewriter */
#define ETXTBSY     26  /* Text file busy */
#define EFBIG       27  /* File too large */
#define ENOSPC      28  /* No space left on device */
#define ESPIPE      29  /* Illegal seek */
#define EROFS       30  /* Read-only file system */
#define EMLINK      31  /* Too many links */
#define EPIPE       32  /* Broken pipe */
#define EDOM        33  /* Math argument out of domain of func */
#define ERANGE      34  /* Math result not representable */

#endif
```

1) Defined as integers
2) Stored in **errno** when error occurs
3) Accessible through "**man errno**"

# Ex) Handling Errors

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <errno.h>

int main() {
    FILE *fp;
    fp = fopen("file.txt", "r");
    if(fp == NULL){
        fprintf(stderr, “Error: %s (%d)\n", errno);
        perror("Error");
        exit(1);
    }
    fclose(fp);
    exit(0);
}
```

```
$ ./test
Error: No such file or directory (2)
Error: No such file or directory
```

# Summary

- **Unix file I/O**
  - **`open()`, `read()`, `write()`, `close()`, `...`**
  - A uniform way to access files, I/O devices, network sockets, kernel data structures, Etc.

- **When to use standard I/O**
  - When working with disk or terminal files

- **When to use raw Unix I/O**
  - When you need to fetch file metadata
  - When you read or write network sockets or pipes
  - In rare cases, when you need absolute highest performance

# Review

- 6 System calls
  - `open()`
  - `close()`
  - `read()`
  - `write()`
  - `lseek()`
  - `stat() / fstat()`

# Example Code - File I/O

```
char filename[] = "hello-dos.txt";
int fd;
char buffer[16];

fd = open(filename, O_RDWR | O_CREAT, 0755);
read(fd, buffer, 6);
read(fd, buffer+6, 2);

lseek(fd, -2, SEEK_CUR);
buffer[0] = '\n';
write(fd, buffer, 1);

lseek(fd, 8, SEEK_SET);
strcpy(buffer, "How");
write(fd, buffer, 3);

close(fd);

fd = open(filename, O_WRONLY | O_CREAT | O_EXCL, 0755);
if (fd < 0)
    printf("errno : %d, error code - EEXIST : %d\n", errno, EEXIST);
```

```
File state (FD: 3)

path: "hello-dos.txt"
position: 0
size: 20
```

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| H | e | l | l | o | . | \r | \n |
| W | h | o | | a | r | e | |
| y | o | u | ? | | | | |

# Example Code - File I/O

```
char filename[] = "hello-dos.txt";
int fd;
char buffer[16];

fd = open(filename, O_RDWR | O_CREAT, 0755);
read(fd, buffer, 6);      // "Hello."
read(fd, buffer+6, 2);

lseek(fd, -2, SEEK_CUR);
buffer[0] = '\n';
write(fd, buffer, 1);

lseek(fd, 8, SEEK_SET);
strcpy(buffer, "How");
write(fd, buffer, 3);

close(fd);

fd = open(filename, O_WRONLY | O_CREAT | O_EXCL, 0755);
if (fd < 0)
    printf("errno : %d, error code - EEXIST : %d\n", errno, EEXIST);
```

```
File state (FD: 3)

path: "hello-dos.txt"
position: 6
size: 20
```

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| H | e | l | l | o | . | \r | \n |
| W | h | o | | a | r | e | |
| y | o | u | ? | | | | |

# Example Code - File I/O

```
char filename[] = "hello-dos.txt";
int fd;
char buffer[16];

fd = open(filename, O_RDWR | O_CREAT, 0755);
read(fd, buffer, 6);
read(fd, buffer+6, 2);  // "Hello.\r\n"

lseek(fd, -2, SEEK_CUR);
buffer[0] = '\n';
write(fd, buffer, 1);

lseek(fd, 8, SEEK_SET);
strcpy(buffer, "How");
write(fd, buffer, 3);

close(fd);

fd = open(filename, O_WRONLY | O_CREAT | O_EXCL, 0755);
if (fd < 0)
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```

```
File state (FD: 3)

path: "hello-dos.txt"
position: 8
size: 20
```

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| H | e | l | l | o | . | \r | \n |
| W | h | o | | a | r | e | |
| y | o | u | ? | | | | |

# Example Code - File I/O

```
char filename[] = "hello-dos.txt";
int fd;
char buffer[16];

fd = open(filename, O_RDWR | O_CREAT, 0755);
read(fd, buffer, 6);
read(fd, buffer+6, 2);

lseek(fd, -2, SEEK_CUR);
buffer[0] = '\n';
write(fd, buffer, 1);

lseek(fd, 8, SEEK_SET);
strcpy(buffer, "How");
write(fd, buffer, 3);

close(fd);

fd = open(filename, O_WRONLY | O_CREAT | O_EXCL, 0755);
if (fd < 0)
    printf("errno : %d, error code - EEXIST : %d\n", errno, EEXIST);
```

```
File state (FD: 3)

path: "hello-dos.txt"
position: 6
size: 20
```

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| H | e | l | l | o | . | \r | \n |
| W | h | o | | a | r | e | |
| y | o | u | ? | | | | |

# Example Code - File I/O

```
char filename[] = "hello-dos.txt";
int fd;
char buffer[16];

fd = open(filename, O_RDWR | O_CREAT, 0755);
read(fd, buffer, 6);
read(fd, buffer+6, 2);

lseek(fd, -2, SEEK_CUR);
buffer[0] = '\n';
write(fd, buffer, 1);

lseek(fd, 8, SEEK_SET);
strcpy(buffer, "How");
write(fd, buffer, 3);

close(fd);

fd = open(filename, O_WRONLY | O_CREAT | O_EXCL, 0755);
if (fd < 0)
    printf("errno : %d, error code - EEXIST : %d\n", errno, EEXIST);
```

```
File state (FD: 3)

path: "hello-dos.txt"
position: 7
size: 20
```

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| H | e | l | l | o | . | \n | \n |
| W | h | o | | a | r | e | |
| y | o | u | ? | | | | |

# Example Code - File I/O

```
char filename[] = "hello-dos.txt";
int fd;
char buffer[16];

fd = open(filename, O_RDWR | O_CREAT, 0755);
read(fd, buffer, 6);
read(fd, buffer+6, 2);

lseek(fd, -2, SEEK_CUR);
buffer[0] = '\n';
write(fd, buffer, 1);

lseek(fd, 8, SEEK_SET);
strcpy(buffer, "How");
write(fd, buffer, 3);

close(fd);

fd = open(filename, O_WRONLY | O_CREAT | O_EXCL, 0755);
if (fd < 0)
    printf("errno : %d, error code - EEXIST : %d\n", errno, EEXIST);
```

```
File state (FD: 3)

path: "hello-dos.txt"
position: 8
size: 20
```

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| h | e | l | l | o | . | \n | \n |
| W | h | o | | a | r | e | |
| y | o | u | ? | | | | |

# Example Code - File I/O

```
char filename[] = "hello-dos.txt";
int fd;
char buffer[16];

fd = open(filename, O_RDWR | O_CREAT, 0755);
read(fd, buffer, 6);
read(fd, buffer+6, 2);

lseek(fd, -2, SEEK_CUR);
buffer[0] = '\n';
write(fd, buffer, 1);

lseek(fd, 8, SEEK_SET);
strcpy(buffer, "How");
write(fd, buffer, 3);

close(fd);

fd = open(filename, O_WRONLY | O_CREAT | O_EXCL, 0755);
if (fd < 0)
    printf("errno : %d, error code - EEXIST : %d\n", errno, EEXIST);
```

```
File state (FD: 3)

path: "hello-dos.txt"
position: 11
size: 20
```

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| H | e | l | l | o | . | \n | \n |
| H | o | w | | a | r | e | |
| y | o | u | ? | | | | |

# Example Code - File I/O

```
char filename[] = "hello-dos.txt";
int fd;
char buffer[16];

fd = open(filename, O_RDWR | O_CREAT, 0755);
read(fd, buffer, 6);
read(fd, buffer+6, 2);

lseek(fd, -2, SEEK_CUR);
buffer[0] = '\n';
write(fd, buffer, 1);

lseek(fd, 8, SEEK_SET);
strcpy(buffer, "How");
write(fd, buffer, 3);

close(fd);

fd = open(filename, O_WRONLY | O_CREAT | O_EXCL, 0755);
if (fd < 0)
    printf("errno : %d, error code - EEXIST : %d\n", errno, EEXIST);
```

```
File state (FD: 3)


:CLOSED
```

# Example Code - File I/O

```
char filename[] = "hello-dos.txt";
int fd;
char buffer[16];

fd = open(filename, O_RDWR | O_CREAT, 0755);
read(fd, buffer, 6);
read(fd, buffer+6, 2);

lseek(fd, -2, SEEK_CUR);
buffer[0] = '\n';
write(fd, buffer, 1);

lseek(fd, 8, SEEK_SET);
strcpy(buffer, "How");
write(fd, buffer, 3);

close(fd);

fd = open(filename, O_WRONLY | O_CREAT | O_EXCL, 0755);
if (fd < 0)
    printf("errno : %d, error code - EEXIST : %d\n", errno, EEXIST);
```

# Lab Exercise

- **Add line numbers using Unix File I/O**
  - Create “week4” directory in your home directory
  - Download "Aladdin.txt" file in your “week4” directory
    - » `$ wget http://www.fpx.de/fp/Disney/Scripts/Aladdin.txt`
  - Create "Aladdin.txt.no" file which has line numbers preceding all text lines
    - » Format: Number Space | Space Text (i.e. transform “line” to “number | line”)
    - » No new line at the end of file, both files should have the same number of lines*
  - The executable receives an argument that is a filename that ends with “.txt”.
    - » You have to use `int main(int argc, char* argv[])`
  - The number of lines can be represented as an unsigned long long integer.

# Lab Exercise

- **Precautions**
  - **Only use Unix File I/O `(open()/close(), read()/write())`**
  - **Don't use other functions like**

    `scanf(), printf(),sprintf(),fscanf(),fprintf(),fopen(),…`

    » **This also applies to string and int-to-str functions (itoa, etc.)**

    » **Otherwise, you may get a zero**
- **Hints**
  - To get "FILENAME.txt.no", get filename length then create array for handling new filename with new length.
  - An unsigned long long integer has up to 20 digits.

    » You can convert a digit to a char by doing `digit + '0'`.
  - If you do not create a file with the right permissions, you might have issues with reading the file.
  - Install dos2unix and then run it on the file

# Lab Exercise

```
ALADDIN:  THE COMPLETE SCRIPT
COMPILED BY BEN SCRIPPS <34RQNPQ@CMUVM.CSV.CMICH.EDU>
(Portions Copyright (c) 1992 The Walt Disney Company)

PEDDLER:    Oh I come from a land
    From a faraway place
    Where the caravan camels roam
    Where they cut off your ear /Where it's flat and immense
    If they don't like your face /And the heat is intense
    It's barbaric, but hey--it's home!
    When the wind's at your back
    And the sun's from the west
    And the sand in the glass is right
    Come on down,
    Stop on by
    Hop a carpet and fly
    To another Arabian night!
```

Aladdin.txt

```
1 | ALADDIN:  THE COMPLETE SCRIPT
2 | COMPILED BY BEN SCRIPPS <34RQNPQ@CMUVM.CSV.CMICH.EDU>
3 | (Portions Copyright (c) 1992 The Walt Disney Company)
4 |
5 | PEDDLER:    Oh I come from a land
6 |     From a faraway place
7 |     Where the caravan camels roam
8 |     Where they cut off your ear /Where it's flat and immense
9 |     If they don't like your face /And the heat is intense
10 |     It's barbaric, but hey--it's home!
11 |     When the wind's at your back
12 |     And the sun's from the west
13 |     And the sand in the glass is right
14 |     Come on down,
15 |     Stop on by
16 |     Hop a carpet and fly
17 |     To another Arabian night!
```

Aladdin.txt.no

```
$ wget http://www.fpx.de/fp/Disney/Scripts/Aladdin.txt
$ sudo apt install dos2unix
$ dos2unix Aladdin.txt
$ ls
Aladdin.txt    w4
$ ./w4 Aladdin.txt
$ ls
Aladdin.txt.no Aladdin.txt    w4
```

# Lab Exercise

- **Submit source code of your exercise**
  - via iCampus
  - Bundle *source code* and *Makefile* with tar command

    ```
    $ tar caf [student_id]_w4.tar.zst week4
    ```

  - We'll grade your submission with **make**

# Lab Exercise

- **Note**
  - The number of lines might differ depending on which vim is used.
  - To make sure the number of lines are the same, you can run `cat -e Aladdin.txt.no`.

```
$
ALADDIN:    A whole new world$
JASMINE:        A whole new life$
BOTH:   (with off-camera chorus)    For you and me!$
MEN'S CHORUS:   A whole new world!$
$
(They fly off into the moonlight, and after they have disappeared,$
    the moon turns and reveals the GENIE's laughing face.  Suddenly$
    the film is grabbed "off the projector", the GENIE lifts it up$
     and looks at the audience.)$
$
GENIE:  Made ya look!$
$
(Drops the film back to normal, with the normal moon.  Fade to black.$
    The end.)$
```

```
$
ALADDIN:    A whole new world$
JASMINE:        A [illegible] new life$
BOTH:   (with of[illegible]horus)    For y[illegible]
MEN'S CHORUS:   [illegible]rld!$
$
(They fly off into [illegible]have disappeared,$
    the moon turns and [illegible]ghing face.  Suddenly$
    the film is grabbed [illegible]the GENIE lifts it up$
     and looks at t[illegible]
$
GENIE:  Made y[illegible]
$
(Drops the film b[illegible] normal, with the norma[illegible]n.  Fade to black.$
    The end.)$
$
```